DR. VISHWANATH KARAD MIT WORLD PEACE UNIVERSITY, PUNE
SCHOOL OF COMPUTER ENGINEERING AND TECHNOLOGY

CCA 3 — COMPUTER VISION PRACTICAL ASSIGNMENT

Comprehensive Report on Units 3, 4, and 5

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PRN / Roll No:	272251066	**Academic Term:**	Term III / Final Term 2025-2026
Subject:	Computer Vision (CCA 3)	**Faculty:**	Prof. Rahul Inchal
Git Repository:	https://github.com/Yash-Ranpariya/cv-	**Video Link:**	Google Drive Shared Link

Abstract

This project implements an end-to-end practical investigation covering Units 3, 4, and 5 of the Computer Vision MCA curriculum. Unit 3 investigates spatial domain image restoration (Gaussian vs Median filtering on AWGN and Salt-and-Pepper impulse noise) and marker-controlled watershed segmentation. Unit 4 implements Histogram of Oriented Gradients (HOG) and Scale-Invariant Feature Transform (SIFT) keypoint matching under rotation and scaling stress tests. Unit 5 compares classical Viola–Jones AdaBoost Haar Cascades with deep convolutional YOLOv8 object recognition and performs temporal motion estimation using Frame Differencing, GMM (MOG2) background subtraction, and Lucas–Kanade optical flow.

1. Task 1: Image Restoration and Segmentation (Unit 3)

Restoration Findings: Gaussian noise smoothing via Gaussian filtering achieves moderate PSNR improvement (28.82 dB), whereas Median filtering delivers superior performance on Salt-and-Pepper noise (31.31 dB PSNR, 0.8949 SSIM), completely purging impulse noise without smearing edges.

Degradation / Condition	Applied Filter	PSNR (dB)	SSIM	Restoration Outcome
Gaussian Noise (sigma=25)	None (Corrupted)	20.59 dB	0.2859	High-frequency variance loss
Gaussian Noise (sigma=25)	Gaussian (5x5)	28.82 dB	0.7405	Noise smoothed; edges preserved
Gaussian Noise (sigma=25)	Median (5x5)	28.44 dB	0.7022	Good variance suppression
Salt & Pepper Noise (3%)	None (Corrupted)	17.06 dB	0.2805	Extreme impulse spikes
Salt & Pepper Noise (3%)	Gaussian (5x5)	26.67 dB	0.6403	Spreads noise into blotches
Salt & Pepper Noise (3%)	Median (5x5)	31.31 dB	0.8949	Optimal: Impulses rejected

2. Task 2: Feature Extraction and Image Matching (Unit 4)

HOG and SIFT Analysis: HOG extracted 9-bin gradient histograms across normalized 8x8 cells. SIFT detected local scale-space extrema in DoG space and generated 128-dimensional invariant descriptors. Under affine stress testing (35 deg rotation + 0.8x scaling), SIFT maintained a high inlier ratio via RANSAC homography.

Experiment Scenario	P2 Keypoints	Lowe's Matches	RANSAC Inliers	Inlier Ratio (%)
Original Scene Pair	2000	220	111	50.5%
Rotated (35 deg) & Scaled (0.8x)	2000	219	106	48.4%

3. Task 3: Object Recognition (Viola–Jones vs YOLOv8) (Unit 5)

Attribute	Viola–Jones (Haar Cascade)	Ultralytics YOLOv8
Underlying Paradigm	Handcrafted Integral Haar Features	Deep CNN Backbone + PANet Neck
Classifier / Head	AdaBoost Attentional Cascade	Anchor-free Decoupled Detection Head
Supported Classes	Human Frontal Faces Only	80 Real-world Classes (MS COCO)
Confidence Output	Binary (No continuous score)	Calibrated Softmax Probability [0.0, 1.0]
Failure Case Observed	Fails on head tilt > 20 deg / profiles	Confidence drops on heavy occlusion

4. Task 4: Video Motion Estimation (Unit 5)

Motion Method	Output Format	Velocity Vector	Handling of Stopped Objects
Frame Differencing	Binary Mask	No (Magnitude only)	Lost immediately (Ghosting)
GMM Background (MOG2)	Probabilistic Mask	No (Presence only)	Retained in background model
Lucas–Kanade Flow	Sparse Vectors	Yes (Direction + Speed)	Vectors decay to zero
Dense Flow (Farneback)	Continuous HSV Field	Yes (Pixel-wise vector)	Velocity magnitude decays

5. Submission Checklist & Verification

- **Original Dataset Files:** Place original_photo.jpg, second_photo.jpg, original_video.mp4 in dataset/.
- **Execution:** Run 'python main.py' to generate all figures in outputs/.
- **Git Push:** Push source code to GitHub repository and paste link into report.
- **Video Demo:** Record 5–8 minute explanation video, upload to Google Drive, and share with rahul.inchal.inchal24@gmail.com and rahul.inchal.vf@mitwpu.edu.in.